

Predicting NHL Playoff Qualification Using Ensemble and Machine Learning

Terah Skillings
Data Science
Master's of Data Science
Regis University, Denver, CO, USA
tskillings@regis.edu

Abstract

This practicum project uses ensemble and machine learning methods to predict which NHL teams will make playoffs using historical NHL data from the 2008 season to 2024. It can be categorized by team performance, roster investment and player availability. Focusing on early season indicators, the analysis used the first twenty games of each season. This project follows the data science workflow, including data cleaning, exploratory data analysis, feature engineering and model development. Multiple machine learning models were used and optimized. These models included both traditional machine learning models to ensemble models such as XGBoost and stacking. Model performance was evaluated on accuracy, ROC-AUC, F1 score, and cross-validation. Results showed the stacking model to have the overall best performance achieving an ROC-AUC score of 0.84. Feature importance showed that goal differential and points are key predictors. Using historical NHL data, this project demonstrates how data science methods can be applied to better understand team performance and playoff outcomes in the NHL.

I. INTRODUCTION/BACKGROUND

Professional sports teams are beginning to use data-driven insights to analyze team performance, individual players and support decision making. Success in the NHL is dependent on how a team is performing, player availability, and coaching changes. As advanced statistics continue to grow in sports, machine learning methods can be used to analyze these variables and identify patterns associated with winning teams. This project uses historical NHL game data to analyze how team performance and roster related variables relate to post season qualification. The main goal is to explore whether machine learning models can predict which teams will make the playoffs and what factors are most valuable for getting there.

II. PROBLEM STATEMENT

This project uses machine learning to predict which NHL teams will make the playoffs based on historical data. Since qualifying for the postseason is dependent on many variables basic statistics are not sufficient to get the full picture. Along with making predictions, this project identifies which specific variables are highly associated with playoff qualification. These insights help explain team performance and provide a better understanding of the overall dynamics of an NHL season.

III. METHODOLOGY/PROPOSED APPROACH

This project follows a structured data science workflow, beginning with raw NHL game-level data from 2008 to 2024. The dataset initially had over 40,000 game level rows and more than 1,000 features. Focusing on early-season indicators, the data was filtered to include only the first 20 games of each season. During data preparation, variables that could lead to data leakage, such as final standings and rankings, were removed. The data was then aggregated to the team-season level, resulting in 522 observations. Feature engineering was applied to create variables such as goal differential, points per game, roster size, and injury rate. Feature selection was then performed by removing low variance and high correlated

variables (correlation > 0.9), resulting in a final dataset of 18 features. Multiple machine learning models were developed and compared, including baseline models and ensemble methods. The baseline models used were: Random Forrest, SVM, Logistic Regression, Naïve Bayes, Decision Tree, Neural Network, and KNN. More advanced models, such as XGBoost and stacking, were created to improve predictive performance. Model evaluation was conducted using accuracy, ROC-AUC, F1 score, and cross-validation. This was done to ensure model reliability and stability.

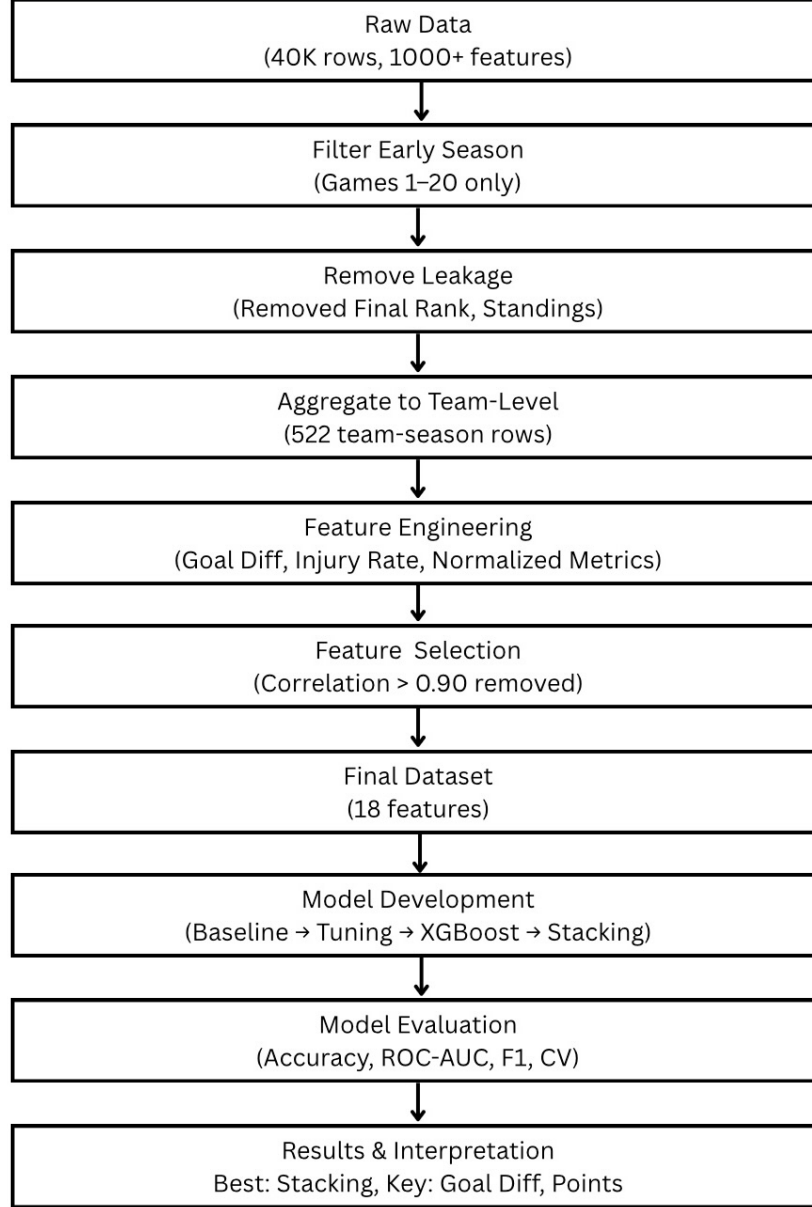


Fig. 1. Data science workflow for predicting NHL playoff qualification, including data preparation, feature engineering, model development, and evaluation.

IV. DATA DESCRIPTION

The NHL dataset contained performance, roster investment, and player availability statistics. The underlying data is publicly available from sources such as MoneyPuck and Naturalstattrick; however, due to the data collection and scraping process performed by a former employer, the compiled dataset used in this project is not publicly distributed. All data was handled locally. Version control was conducted

in Jupyter notebooks to track week to week progress and changes. Aggregation to the team-season helped create an early-season story line for that team rather than game by game. Feature selection techniques were used to remove redundant variables, and highly correlated features were eliminated to improve model stability. Ethically this data is public and should not raise any privacy or ethical concerns. Various models were implemented in order to ensure the results were reliable and stable. This project follows responsible data science practices by emphasizing transparency, reproducibility, and appropriate handling of data throughout the analysis process.

V. RESULTS AND DISCUSSION

Multiple machine learning models were implemented and evaluated to predict NHL playoff qualification using the first 20 games played in that NHL season. Baseline models showed relatively consistent performance across models. These untuned models had accuracy values ranging from approximately 0.66 to 0.71 and ROC-AUC scores between 0.71 and 0.8. Naive Bayes achieved the highest ROC-AUC of 0.8, followed by Logistic Regression at 0.782. Random Forest and Support Vector Machine (SVM) also performed well, with ROC-AUC scores of 0.767 and 0.773 respectively. These results suggest that both linear and nonlinear relationships are present in the data and can be captured by various models. Although Naïve Bayes achieved the highest ROC-AUC among baseline models, it was not selected for further modeling due to the lower performance in cross-validation. To further improve the models performance

Model	Accuracy	ROC AUC	F1	CV Score
Decision Tree	0.676	0.717	0.685	0.722
KNN	0.667	0.737	0.667	0.746
Logistic Regression	0.686	0.782	0.692	0.787
Naïve Bayes	0.714	0.8	0.712	0.77
Neural Network	0.695	0.72	0.704	0.731
Random Forrest	0.705	0.767	0.71	0.779
Stacking	0.771	0.84	0.8	0.834
SVM	0.705	0.773	0.71	0.789
XGBoost	0.752	0.84	0.768	0.797

Fig. 2. Performance comparison of baseline machine learning models for NHL playoff prediction using accuracy, ROC-AUC, F1 score, and cross-validation.

hyperparameter tuning was applied to the most balanced overall top performing models, Random Forest, SVM, and Logistic Regression. The tuning did not result in significant improvements. The tuned Random Forest, SVM, and Logistic Regression all achieved a ROC-AUC of 0.78, with only minor changes in accuracy and F1 scores. This signifies that model performance is more dependent on the quality of the features and data representation rather than hyperparameter optimization alone.

More advanced models were then implemented to explore predictive performance. The XGBoost model achieved an accuracy of 0.75, ROC-AUC of 0.84, and F1 score of 0.77, with a cross-validation score of 0.80. The stacking model, which used Random Forest, SVM, and XGBoost as a base, had the best overall performance. It achieved an accuracy of 0.77, ROC-AUC of 0.84, and F1 score of 0.80, with a cross-validation of 0.83. Both XGBoost and stacking achieved similar ROC-AUC values. The stacking model demonstrated slightly better overall balance across evaluation metrics and stronger cross-validation performance, suggesting better generalization. A feature importance analysis from the Random Forest model showed that Goal Differential was the most influential predictor, followed by Points and Goals Against. Goals For and injury-related features, such as Injury Rate and Total Injuries, had moderate importance. This indicates that while roster health plays a role in playoff qualification, it is second to

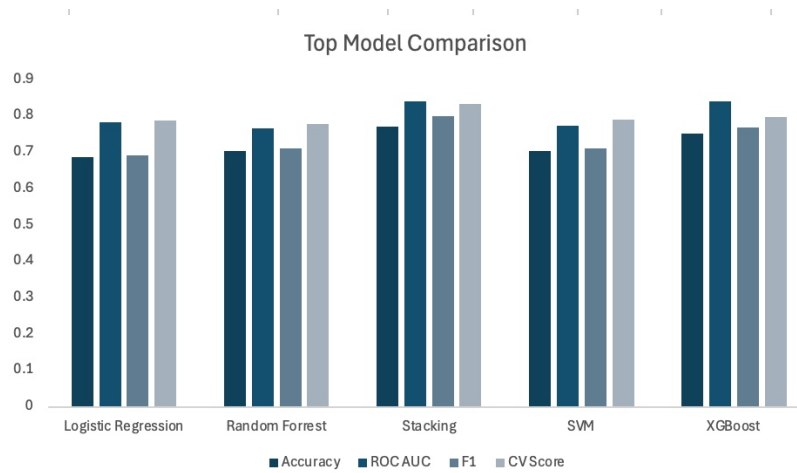


Fig. 3. Comparison of scores across top models, highlighting performance improvements from baseline to ensemble methods..

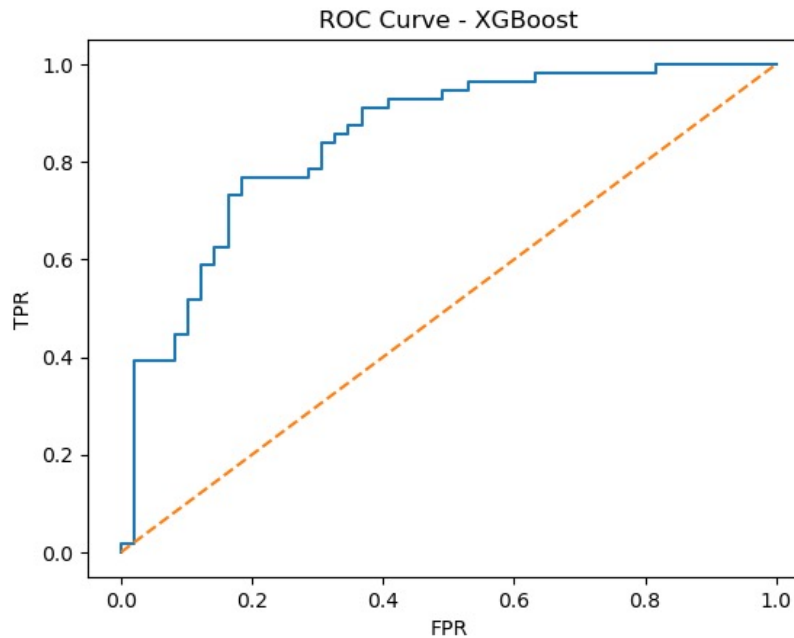


Fig. 4. ROC curve for the stacking model, illustrating strong classification performance across thresholds.

overall team performance. Additional features such as roster composition and player cost variables had smaller contributions. This suggests that their impact is more indirect. The results show that early-season performance metrics are strong indicators of playoff qualification. Ensemble methods, specifically stacking, improved predictive performance compared to individual models by combining multiple approaches. The models achieved strong results being limited to the first 20 games of the season. The area of uncertainty is a reason sports are popular because they do have some unpredictability. Despite this limitation, the models provide meaningful insight into the key factors associated with NHL playoff success.

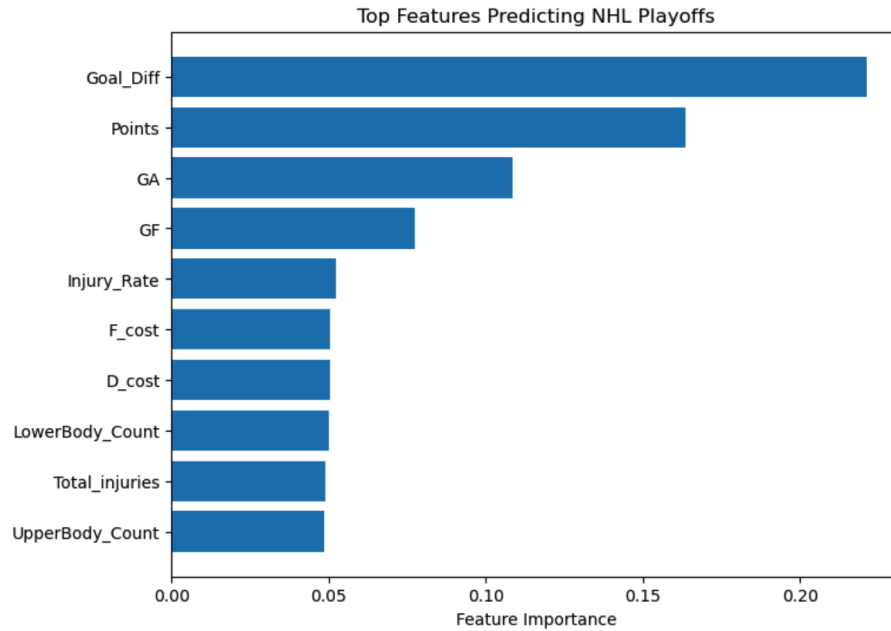


Fig. 5. Feature importance ranking showing the most influential variables in predicting NHL playoff qualification.

VI. TIMELINE

Weeks 1–2	Project planning, proposal development, and familiarization with dataset.
Week 3	Continue examining dataset and begin data cleaning. Work on understanding potential models such as Stacking
Week 4	Conduct exploratory data analysis (EDA) and feature engineering
Week 5	Run baseline machine learning models
Week 6	Tune machine learning and implement XGBoost and stacking model
Week 7	Complete final revisions and prepare deliverables.
Week 8	Final project presentation.

VII. CONCLUSION

This project used machine learning to predict which NHL teams would qualify for the playoffs based on early-season data. The project focused on the first 20 games of each season (2008-2024) to see if early signs of team performance and roster health could accurately predict postseason outcomes. The traditional machine learning models were relatively successful, but the ensemble models gave the strongest results. Stacking performed the best, being very accurate (ROC-AUC of about 0.84), showing that combining multiple models is effective. Random Forrest also found the key factors for playoff success. The most important were goal differential and points, confirming that overall team performance matters most. Injury-related features played a smaller role. This project demonstrates how data science methods can be used to solve real sports analytics problems by combining hockey knowledge with strategic modeling. The model only used early-season data, but it provides a strong foundation for future work such as predicting Stanley Cup winners or how far into playoffs a team will go.